

Composite, inverse and periodic functions

Three ways one function can be built from another — chaining it, undoing it, and repeating it.

LESSON 17–18

2 × 40 MIN

ALGEBRA 10, §1.5

P1 · 2.4–2.6

LESSON CLOCK

10'

12'

16'

14'

8'

5'

Composition — inner first	10 min
fg is not gf	12 min
Inverses	16 min
Periodic functions	14 min
Practice	8 min
Homework	5 min

LEARNING OBJECTIVES

- Form and evaluate the composite functions fg and gf .
- Show that $fg \neq gf$ in general.
- Find the inverse of a one-to-one function and state its domain and range.
- Recognise a periodic function and state its period.

TEXTBOOK

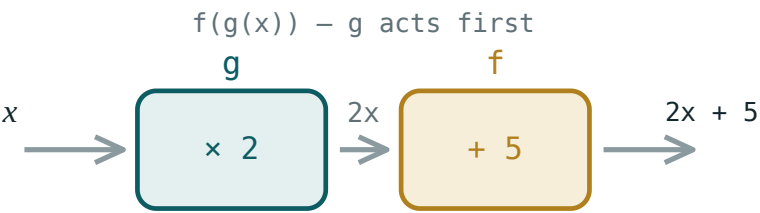
National	pp. 50–58
Cambridge	Pure Mathematics 1, pp. 42–55
Workbook	P1 Exercise 2D, 2E

01

Explanation

Composition — the inner function goes first

$fg(x)$ means $f(g(x))$: apply g , then feed the result into f . The function written **closest to the x** acts first.



Two machines in a line. The output of g becomes the input of f .

$f(x) = x^2, g(x) = x + 3 \Rightarrow fg(x) = (x + 3)^2 \text{ but } gf(x) = x^2 + 3$

ORDER MATTERS

fg and gf are different functions. They agree only for special pairs — and finding the x where they happen to be equal is a standard exam question.

The domain of fg is the set of x in the domain of g whose image $g(x)$ lies in the domain of f .

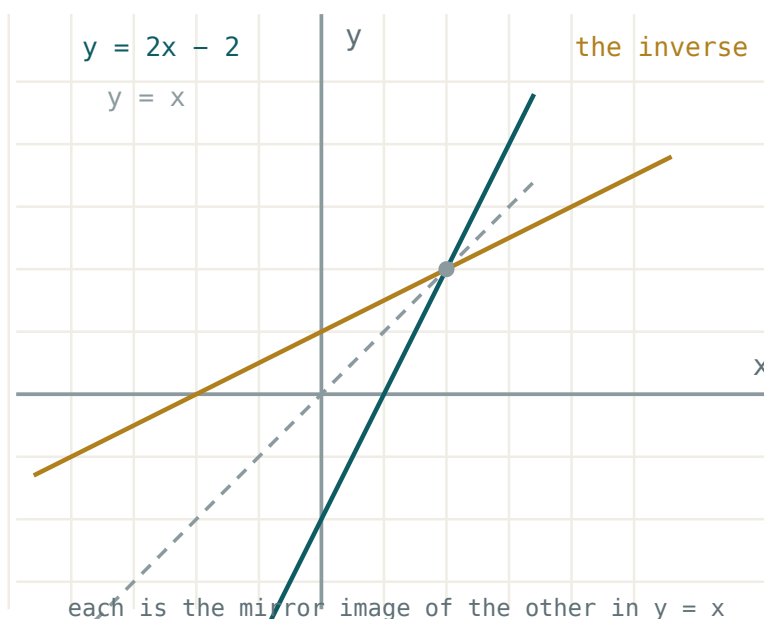
Inverses — undoing the machine

ONLY ONE-TO-ONE FUNCTIONS HAVE INVERSES

If two inputs share an output, the reverse arrow does not know where to go. That is why $f(x) = x^2$ on $\mathbb{R}$ has no inverse, but on $x \geq 0$ it has one.

The method is always the same three steps:

1. Write $y = f(x)$.
2. Make x the subject.
3. Swap the letters — and swap domain with range.



*The inverse is the mirror image in the line $y = x$.
Domain and range swap places.*

$$f(x) = 3x - 5 \Rightarrow y = 3x - 5 \Rightarrow x = \frac{y + 5}{3} \Rightarrow f^{-1}(x) = \frac{x + 5}{3}$$

f^{-1} IS NOT $\frac{1}{f}$

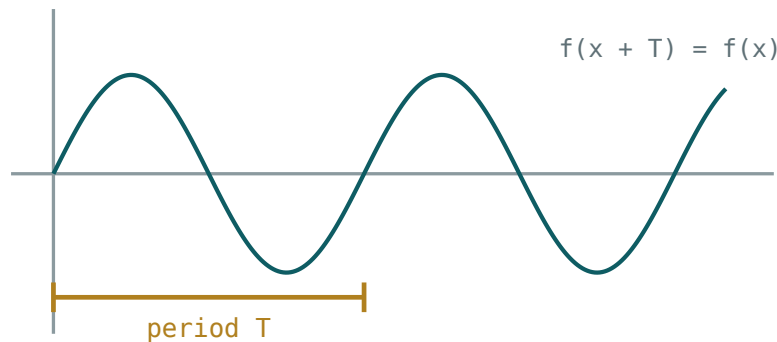
The -1 is notation for “inverse”, not an index. $f^{-1}(x)$ and $[f(x)]^{-1}$ are different things.

The defining property: $f^{-1}f(x) = x$ and $ff^{-1}(x) = x$. Composing a function with its inverse gives the identity — that is the check to run on every answer.

Periodic functions

A function is **periodic** when its whole graph repeats after a fixed horizontal shift:

$f(x + T) = f(x)$ for every x , with the smallest such $T > 0$ called the period



One block of the graph, repeated forever. The period is the width of the block.

Trigonometric functions are the standard examples — $\sin x$ and $\cos x$ have period 360° , $\tan x$ has period 180° . So do many real quantities: hours of daylight, tides, the current in an alternating circuit.

SMALLEST

$\sin(x + 720^\circ) = \sin x$ too — but 720° is not the period, because 360° works and is smaller.

02 Worked examples

EXAMPLE 1 $f(x) = 2x + 1$, $g(x) = x^2$. Find $fg(3)$ and $gf(3)$.

① $g(3) = 9$

Inner first.

② $fg(3) = f(9) = 19$

③ $f(3) = 7$

④ $gf(3) = g(7) = 49$

Different, as expected.

ANSWER

$$fg(3) = 19, gf(3) = 49$$

EXAMPLE 2 Find f^{-1} for $f(x) = \frac{x+2}{x-1}$, $x \neq 1$.

① $y = \frac{x+2}{x-1}$

② $y(x-1) = x+2 \Rightarrow yx - y = x+2$

③ $x(y-1) = y+2 \Rightarrow x = \frac{y+2}{y-1}$

④ Swap letters.

This function is its own inverse.

ANSWER

$$f^{-1}(x) = \frac{x+2}{x-1}, x \neq 1$$

EXAMPLE 3 $f(x) = x + 4$, $g(x) = 3x$. Solve $fg(x) = gf(x)$.

① $fg(x) = 3x + 4$

② $gf(x) = 3(x + 4) = 3x + 12$

③ $3x + 4 = 3x + 12 \Rightarrow 4 = 12$

A contradiction.

ANSWER

No solution — they are never equal

03 Interactive model

Feed a value through both orders and watch the two answers separate.

Inner first, outer second

$f(x) = x + 1$, $g(x) = 2x$. Then $fg(3)$ is:

Same pair; $gf(3)$ is:

The inverse of $f(x) = 5x$ is:

$f(x) = x^2$ on $\mathbb{R}$ has an inverse because:

it does not

it is one-to-one

it is continuous

it is even

The graph of f^{-1} is the graph of f reflected in:

the x-axis

the y-axis

$y = x$

the origin

If $f(x + 6) = f(x)$ for all x , the period could be:

6

12

3

any of these

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Composite function	Murakkab funksiya	Сложная функция
Inner function	Ichki funksiya	Внутренняя функция
Outer function	Tashqi funksiya	Внешняя функция
Inverse function	Teskari funksiya	Обратная функция
Identity function	Ayniy funksiya	Тождественная функция
Periodic function	Davriy funksiya	Периодическая функция
Period	Davr	Период
One-to-one	Biror-birga	Взаимно однозначное
Reflection in $y = x$	$y = x$ ga nisbatan simmetriya	Симметрия относительно $y = x$

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$fg(x)$ means:

$ff^{-1}(x)$ equals:

A function has an inverse when it is:

The period of $\cos x$ is:

90°

180°

360°

720°

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $f = x + 2, g = 3x$; find $fg(1)$

ANSWER 5

2. Same pair; $gf(1)$

ANSWER 9

3. Inverse of $f(x) = x + 7$

ANSWER $x - 7$

4. Inverse of $f(x) = 4x$

ANSWER $\frac{x}{4}$

5. Period of $\sin x$

ANSWER 360°

6. Period of $\tan x$

ANSWER 180°

7. $f^{-1}f(5)$ equals

ANSWER 5

Medium · the standard question

7 PROBLEMS

1. $f = x^2, g = x - 1$; find $fg(x)$

ANSWER $(x - 1)^2$

2. Same pair; $gf(x)$

ANSWER $x^2 - 1$

3. Inverse of $f(x) = 2x - 9$

ANSWER $\frac{x + 9}{2}$

4. Inverse of $f(x) = \frac{1}{x}$

ANSWER $\frac{1}{x}$

5. $f = 3x, g = x + 4$; solve $fg(x) = 0$

ANSWER $x = -4$

6. Inverse of $f(x) = x^3 + 1$

ANSWER $\sqrt[3]{x - 1}$

7. If $f(x + 5) = f(x)$ and no smaller shift works, the period is

ANSWER 5

Hard · stretch and reason

7 PROBLEMS

1. $f = \sqrt{x}, g = x - 4$; domain of fg

ANSWER $x \geq 4$

2. Same pair; domain of gf

ANSWER $x \geq 0$

3. Inverse of $f(x) = \frac{2x + 1}{x - 3}$

ANSWER $\frac{3x + 1}{x - 2}$

4. $f = x + 3, g = x^2$; solve $fg(x) = gf(x)$

ANSWER $x = -1$

5. Restrict $f(x) = x^2 - 4x$ so that it has an inverse

ANSWER $x \geq 2$ (or $x \leq 2$)

6. Find f^{-1} for that restriction

ANSWER $2 + \sqrt{x + 4}$

7. Show that if f has period T then so does $f(x) + 5$

ANSWER Adding a constant does not change the repeat

07 Homework

Homework — 5 tasks

Check every inverse by composing it with the original.

- $f(x) = 3x - 2, g(x) = x^2$. Find $fg(x)$ and $gf(x)$.
- Find f^{-1} for $f(x) = 5 - 2x$ and check $f^{-1}f(4) = 4$.
- Find f^{-1} for $f(x) = \frac{x}{x + 1}, x \neq -1$.
- Explain why $f(x) = x^4$ has no inverse on $\mathbb{R}$, and give a domain where it does.

5. Sketch a periodic function with period 4 that is not trigonometric.